SRM UNIVERSITY–AP

COMPUTER SCIENCE AND ENGINEERING

MACHINE LEARNING PROJECT REPORT

on

"CUSTOMER CHURN PREDICTION USING MACHINE LEARNING"

Submitted in partial fulfillment of the requirements for the course

MACHINE LEARNING

Submitted by

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Academic Year 2026–2027

Abstract

Customer churn is a major challenge for businesses, particularly in telecommunications and subscription-based industries. It refers to customers discontinuing a service or switching to another provider. Predicting churn helps businesses identify customers who may leave and take preventive action to retain them.

This case study applies supervised machine learning to classify customers as likely to churn or likely to stay. The IBM Telco Customer Churn dataset, containing 7,043 customer records and 21 columns, was used. After preprocessing and encoding of categorical variables, four classification algorithms (Logistic Regression, Decision Tree, Random Forest and Gradient Boosting) were trained on 5,634 records and evaluated on 1,409 test records using accuracy, precision, recall, F1-score and confusion matrices. Logistic Regression and Gradient Boosting achieved the highest accuracy (0.806), and Logistic Regression achieved the highest recall (0.559) and F1-score (0.604).

1. Introduction

Customer retention is essential for the growth and success of any business. Losing existing customers reduces revenue and increases the cost of acquiring new ones. Identifying customers who are likely to leave at an early stage allows a business to take appropriate action.

Traditional methods of identifying churn depend mainly on customer feedback and historical reports. Machine learning offers a more effective approach by analyzing customer data and identifying patterns that indicate possible departure.

The objective of this project is to build and compare classification models that predict whether a customer is likely to churn or likely to stay, based on historical customer information. This is an offline prediction study, not a live customer-monitoring service.

2. Dataset Description

The project uses the IBM Telco Customer Churn dataset, which contains customer-related information from a telecommunications service provider. It has 7,043 records and 21 columns covering customer demographics, subscribed services, account information, contract types, payment methods and billing details. The target variable, Churn, has two classes: Yes (the customer left) and No (the customer stayed). The important attributes are listed in Table 1.

Table 1. Important attributes of the dataset.

Attribute	Description
Gender	Customer's gender
SeniorCitizen	Whether the customer is a senior citizen
Partner	Whether the customer has a partner
Tenure	Number of months the customer has stayed with the company
InternetService	Type of internet service subscribed
Contract	Customer's contract type
MonthlyCharges	Monthly amount charged
TotalCharges	Total amount charged during the subscription
PaymentMethod	Customer's payment method
Churn (target)	Yes = customer left, No = customer stayed

3. Methodology

The implementation was carried out in Python using Google Colab, Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn. The workflow included dataset loading, inspection, preprocessing, feature engineering, train–test splitting, model training, evaluation and visualization, as shown in Figure 1.

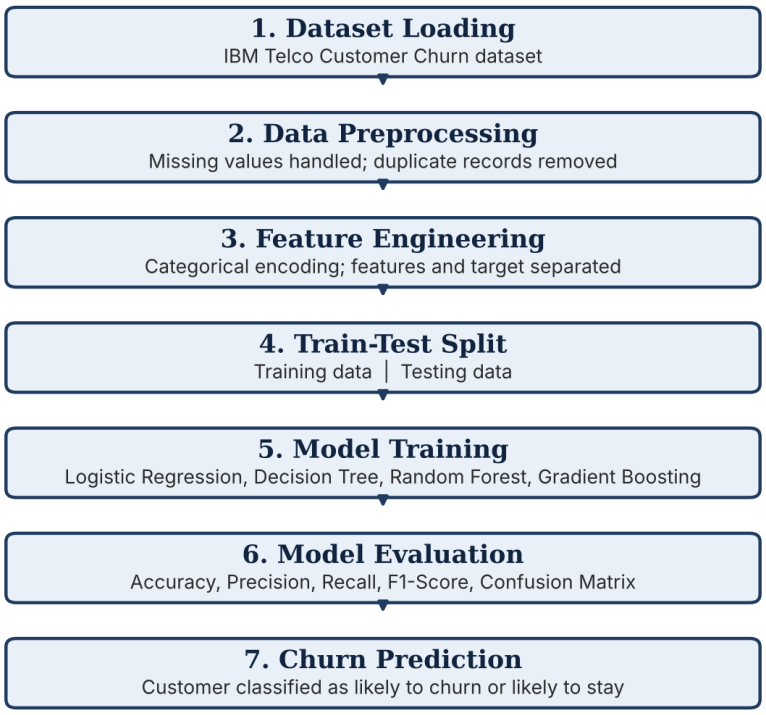


Figure 1. Workflow of Customer Churn Prediction Using Machine Learning.

3.1 Data Preprocessing

- The dataset (7,043 records, 21 columns) was loaded and inspected.
- TotalCharges was converted to a numeric type; any entry that could not be converted was treated as a missing value.
- Numerical missing values were handled using median imputation, and categorical missing values using most-frequent imputation.
- Duplicate records were checked for; none were found (0 removed).
- The customerID column was dropped, as it only identifies a customer.
- Categorical variables were converted into numerical form using One-Hot Encoding, and numerical features were standardized.
- The target Churn was encoded as 1 (Yes) and 0 (No), and the data was divided into 80% training records (5,634) and 20% testing records (1,409) using a stratified split with a fixed random state of 42.

3.2 Model Development

Four supervised classification algorithms were implemented, each inside a pipeline that applies the same preprocessing steps:

1. **Logistic Regression:** predicts the probability of customer churn and is suitable for binary classification.
2. **Decision Tree Classifier:** uses decision rules to classify customers based on their characteristics.
3. **Random Forest Classifier:** combines multiple decision trees to improve prediction performance and reduce overfitting.
4. **Gradient Boosting Classifier:** builds models sequentially, with each model attempting to correct the errors of the previous ones.

3.3 Evaluation Metrics

The models were compared on the test set using the following metrics, with churn (Yes) treated as the positive class:

- **Accuracy:** the proportion of all predictions that are correct.
- **Precision:** how many customers predicted to churn actually belong to the churn class.
- **Recall:** how many actual churn customers are correctly identified.
- **F1-Score:** the harmonic balance between precision and recall.
- **Confusion Matrix:** a table of correct and incorrect classifications.

Higher values of all four scores indicate better classification performance.

4. Implementation and Visualization

The project was implemented in Python in a Google Colab notebook using the following libraries: Pandas for dataset handling, NumPy for numerical calculations, Matplotlib for visualization, Seaborn for statistical graphs and heatmaps, and Scikit-learn for preprocessing, model training and evaluation.

The notebook generated the churn distribution (Figure 2), the correlation heatmap of numerical features and churn (Figure 3), a comparison chart of the evaluation metrics for all four models (Figure 4) and the confusion matrix of each model (Figure 5). These charts were considered together with the numerical evaluation metrics.

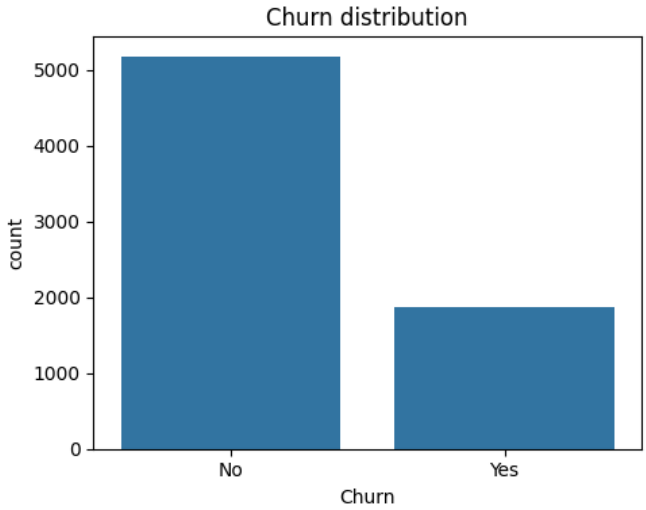


Figure 2. Customer churn distribution.

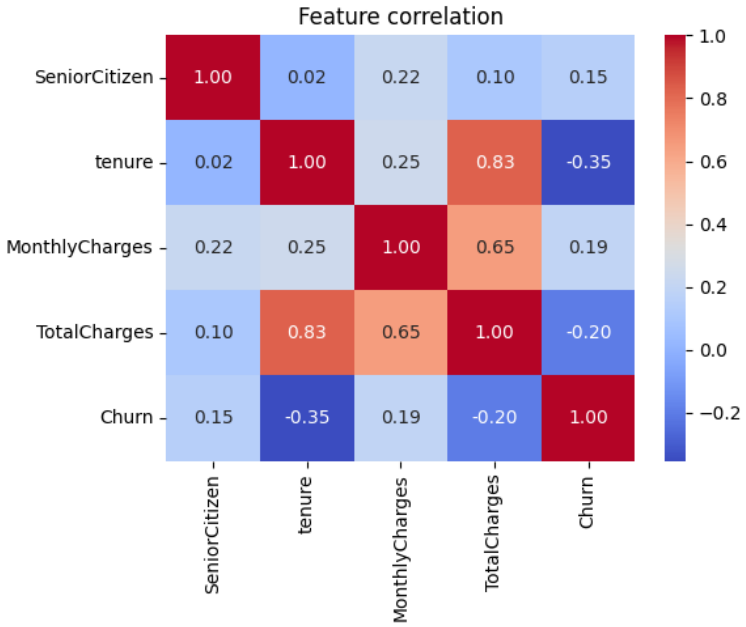


Figure 3. Correlation heatmap of numerical features and churn.

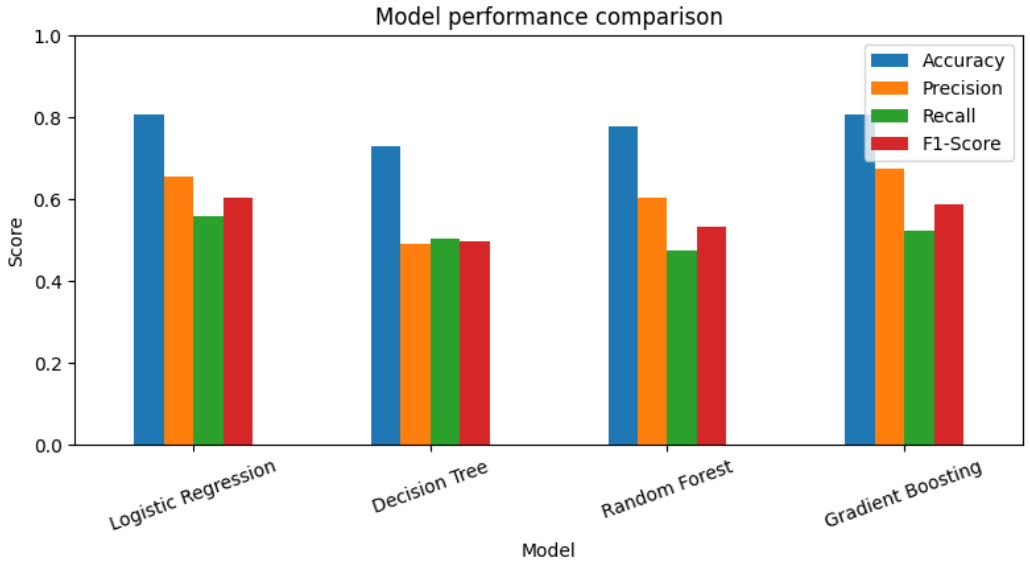


Figure 4. Performance comparison of the four models.

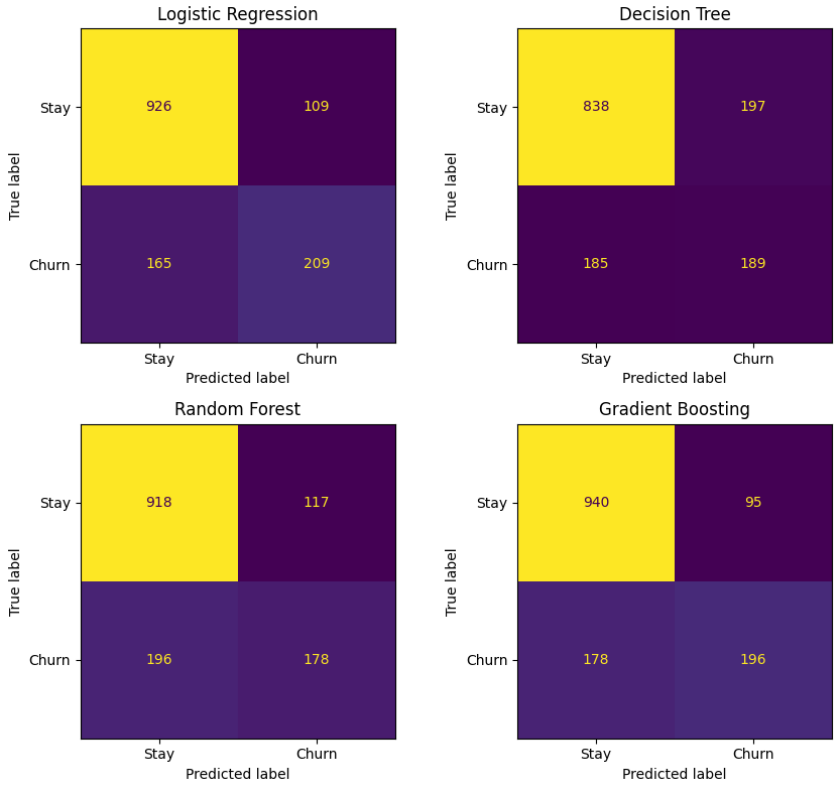


Figure 5. Confusion matrices of the four models.

5. Results and Discussion

The trained models were evaluated on the 1,409 test records. Table 2 shows the values obtained; the best value in each column is shown in bold.

Table 2. Test-set performance of the four models.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	**0.806**	0.657	**0.559**	**0.604**
Decision Tree	0.729	0.490	0.505	0.497
Random Forest	0.778	0.603	0.476	0.532
Gradient Boosting	**0.806**	**0.674**	0.524	0.589

5.1 Discussion

Logistic Regression and Gradient Boosting reached the same highest accuracy of 0.806. Logistic Regression had the highest recall (0.559) and F1-score (0.604), meaning it identified the largest share of customers who actually left while keeping a good balance with precision. Gradient Boosting had the highest precision (0.674), meaning that when it predicted churn it was correct more often than the other models.

The Decision Tree was the weakest model, with the lowest accuracy (0.729), precision (0.490) and F1-score (0.497). Random Forest achieved an accuracy of 0.778 but the lowest recall (0.476), so it missed more churn customers than any other model. Accuracy alone can therefore be misleading for this problem, because customers who stayed outnumber customers who left (Figure 2); precision, recall and the confusion matrices give a fuller picture.

In churn prediction, a high recall allows retention action to reach most customers who are about to leave, while a high precision avoids spending effort on customers who would have stayed. Considering both, Logistic Regression gives the best overall balance in this study, with Gradient Boosting as a close alternative when fewer false alarms are preferred. Even the best model identified only about 56% of the churn customers, so a substantial share still goes undetected. These results come from a single stratified train–test split, so small differences between the two leading models should be interpreted with caution.

6. Conclusion and Future Work

This project demonstrated how machine learning can be used to predict customer churn by analyzing historical customer information. Four classifiers (Logistic Regression, Decision Tree, Random Forest and Gradient Boosting) were trained and compared on the IBM Telco Customer Churn dataset. Logistic Regression achieved the best balance of precision and recall (recall 0.559, F1-score 0.604, accuracy 0.806), while Gradient Boosting matched its accuracy and had the highest precision (0.674). The approach can help businesses identify customers at risk of leaving, improve retention strategies and make better business decisions.

The study is an offline analysis of historical data; it does not include live data or automated retention actions, and it does not analyze which features influence churn most.

Future work can include:

- Integrating real-time customer data for continuous prediction.
- Developing a dashboard to display churn predictions.
- Applying advanced machine learning and deep learning algorithms.
- Identifying the major factors influencing customer churn.
- Providing personalized customer retention recommendations.
- Generating automated alerts for high-risk customers.

References

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